Coding & Solution

Date	03 july 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Harshagv123/OptiCrop
Programming Language(s)	Python, HTML
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	Crop Recommendation System, Data Preprocessing, Machine Learning Model Training, Crop Prediction, Flask Web Application, Interactive User Interface
Pending / Incomplete Features	Real-time Weather Integration, Fertilizer Recommendation Module, Mobile Application Support
Setup / Run Instructions	Install required dependencies using pip install -r requirements.txt, activate virtual environment, and run the application using python app.py

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop application successfully implements a machine learning-based crop recommendation system using soil and environmental parameters. The project follows a modular architecture with separate components for data preprocessing, model training, prediction, and web application development. The solution meets the primary project objectives by providing accurate crop recommendations through a user-friendly Flask interface. Future enhancements may include weather API integration, advanced analytics dashboards, and fertilizer recommendation features.